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Multimodal Techniques for Financial Stock Forecasting

CAPSTONE PRESENTATION

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Problem Statement

- Financial markets react to numbers and narratives.
- Traditional forecasting models rely heavily on historical price data but this approach often fails to capture contextual signals in financial news and company reports.
- Moreover, there is no clear consensus in existing research on
  - ★ Which data modalities contribute most to prediction performance
  - ★ How different modalities should be effectively combined
- To address this gap, we conduct a comparative study to evaluate their impact on a 5 day stock price forecasting task.

Dataset - FinMultiTime

All experiments use the FinMultiTime dataset. We construct a reduced, task specific version by excluding image data and HS300 samples.

Coverage - S&P 500 constituent stocks over 2009–2025

Modalities:

- Time Series - Daily OHLCV (Open, High, Low, Close Volume) records per stock
- Financial News - Nasdaq articles with LSA summarization
- Financial Reports and Earnings Data - SEC XBRL filings sourced from EDGAR API covering balance sheet, cash flow, and equity statements.

Exploratory Data Analysis

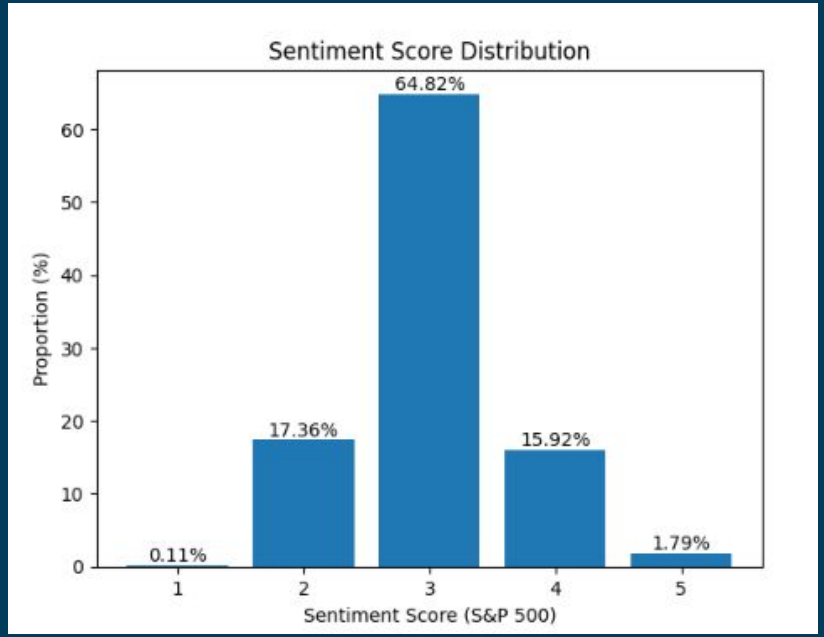


Figure 1: Distribution of sentiment scores in the text modality.

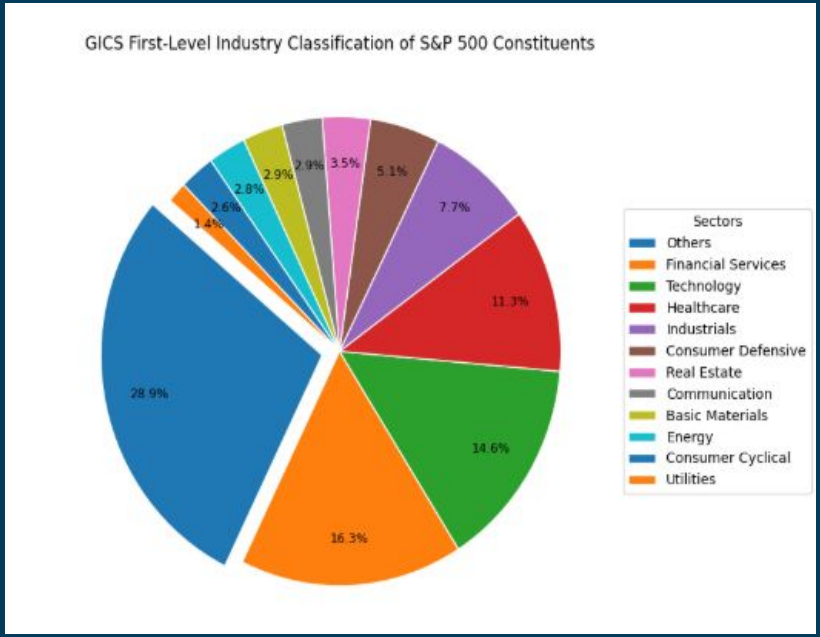
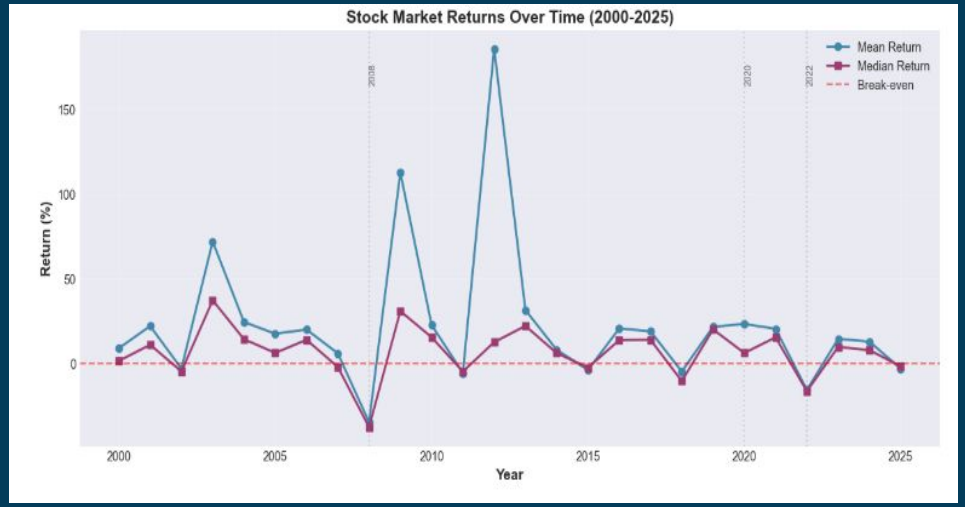
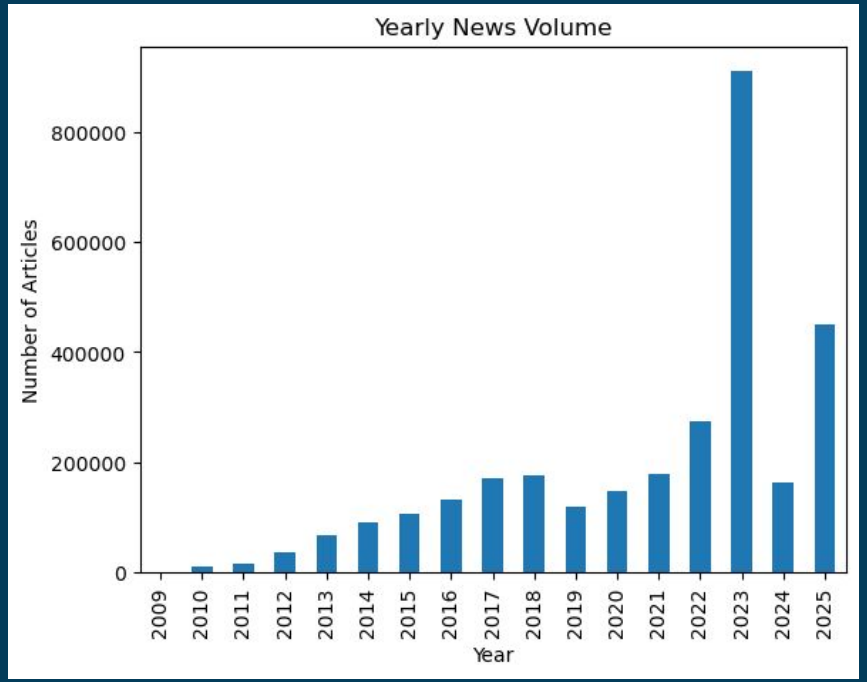


Figure 2: Distribution of stocks across GICS sectors in the relational metadata.





In our temporal EDA we tried to understand the data by looking at both news and time-series data. For our news data we tried to look at the following:-

- Distribution of articles over time
- Other plot shows the yearly stock market return over time (2000-2025)

EXPERIMENTS & FINDINGS

Temporal Fusion Transformer

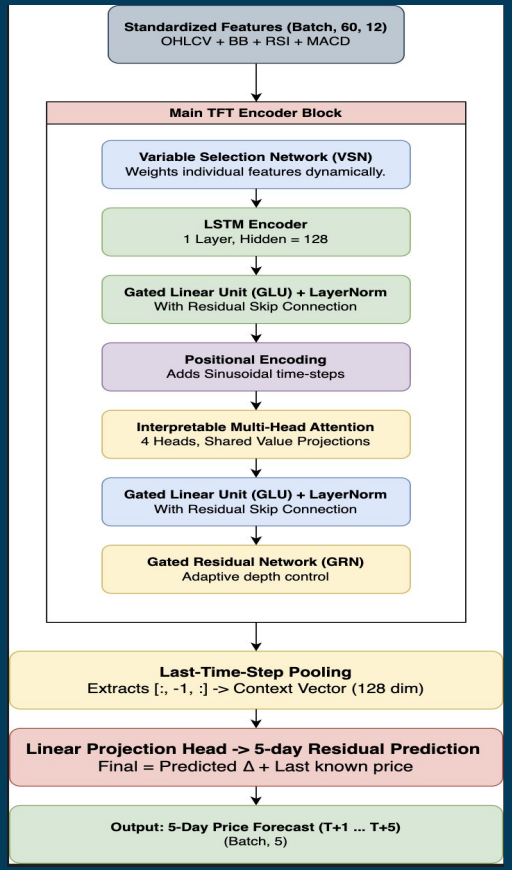
1. **Preprocessing**

a. 5 base features (OHLCV) + 7 technical indicators (Bollinger Bands, RSI, MACD) = 12 features.

2. **Model**

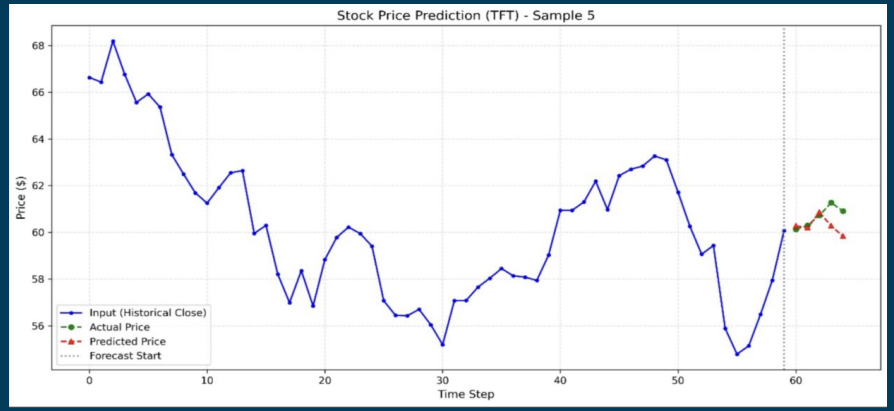
a. Feature Selection + Gated Architecture: Automatically weighs the most relevant features and gates information flow to suppress noise.

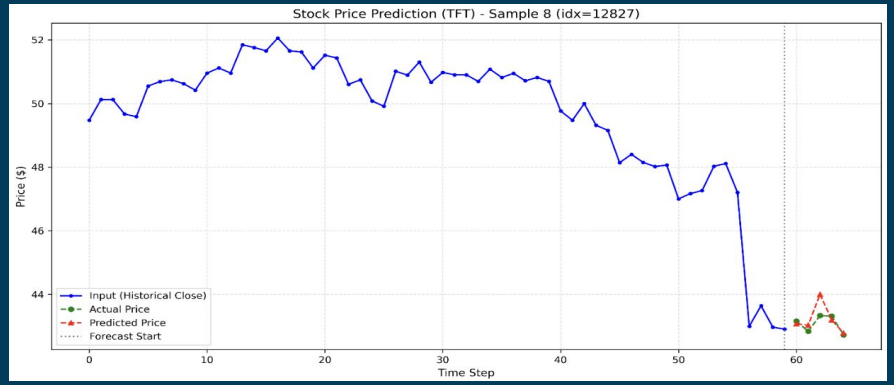
b. LSTM + Attention -> 5 Day Forecast: Combines sequential and Long range patterns to predict a full price trajectory in one pass.



Temporal Fusion Transformer Results

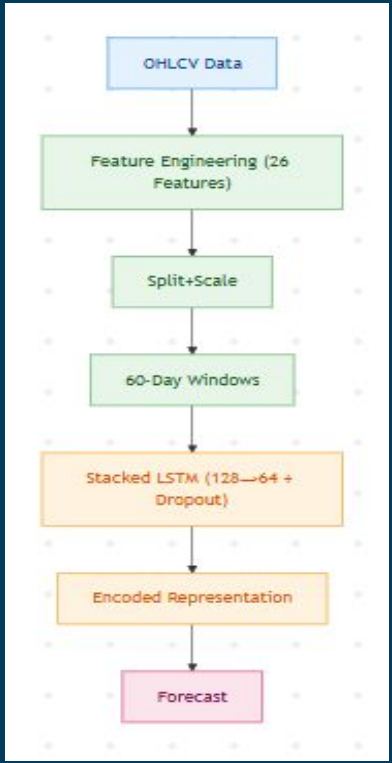
Metric	Value
MAE	0.0358
MSE	0.0046
RMSE	0.0681





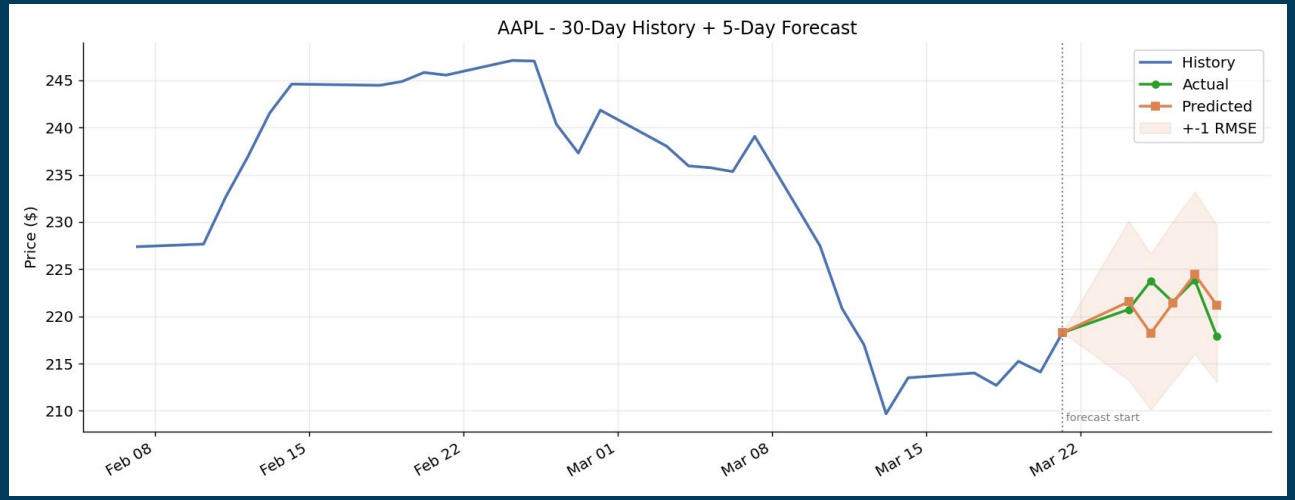
LSTM

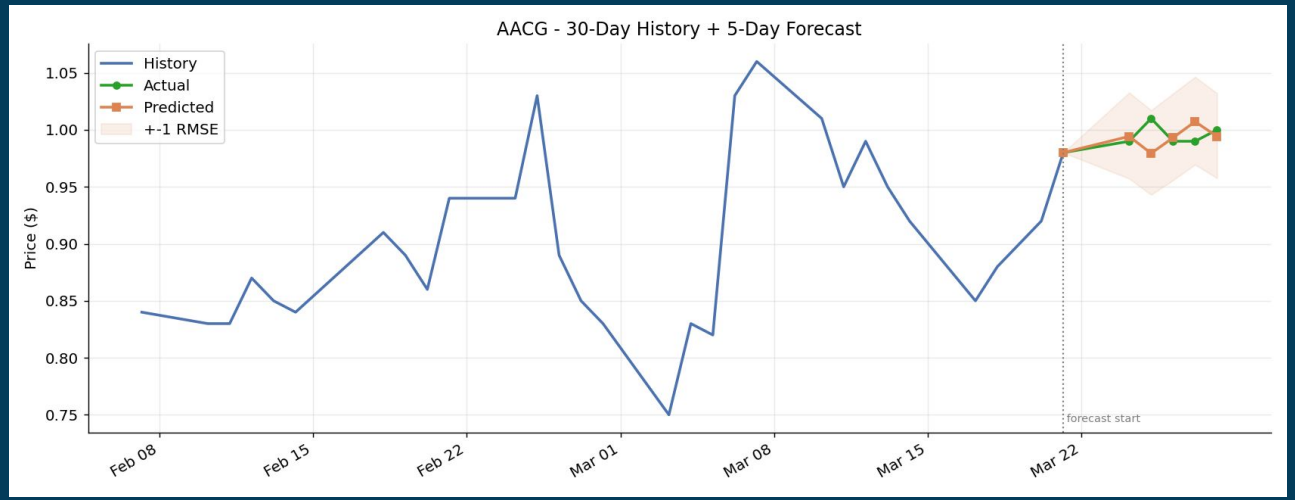
1. Features: Returns, price patterns, volume trends, rolling stats, technical indicators (RSI, ATR, MACD, Bollinger)
2. Input/Output: 60-day window → 5-day forecast
3. Model: Stacked LSTM (128 → 64 units, dropout, ~130K params)
4. Learns multi-scale temporal patterns
5. Direct multi-step forecasting: This avoids recursive predictions and reduces error accumulation.



RESULTS

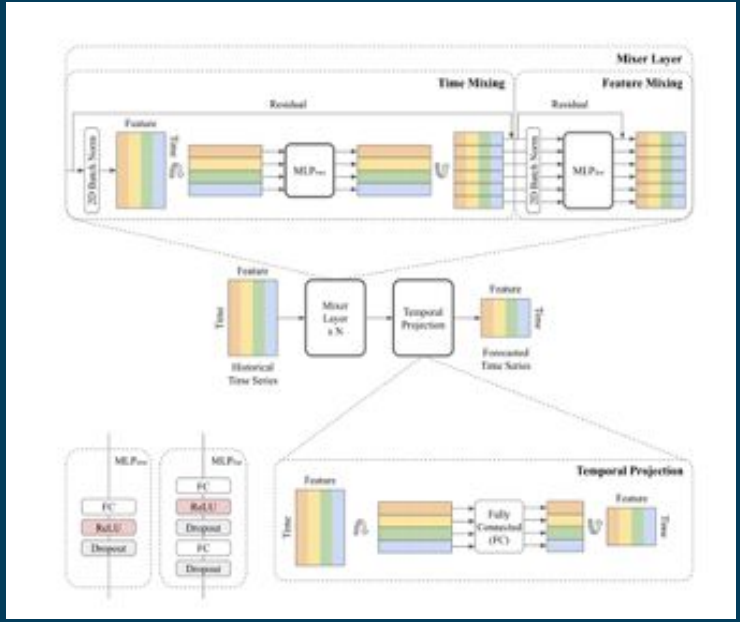
Metric	Value
MAE	0.02939
MSE	0.00143
RMSE	0.03794





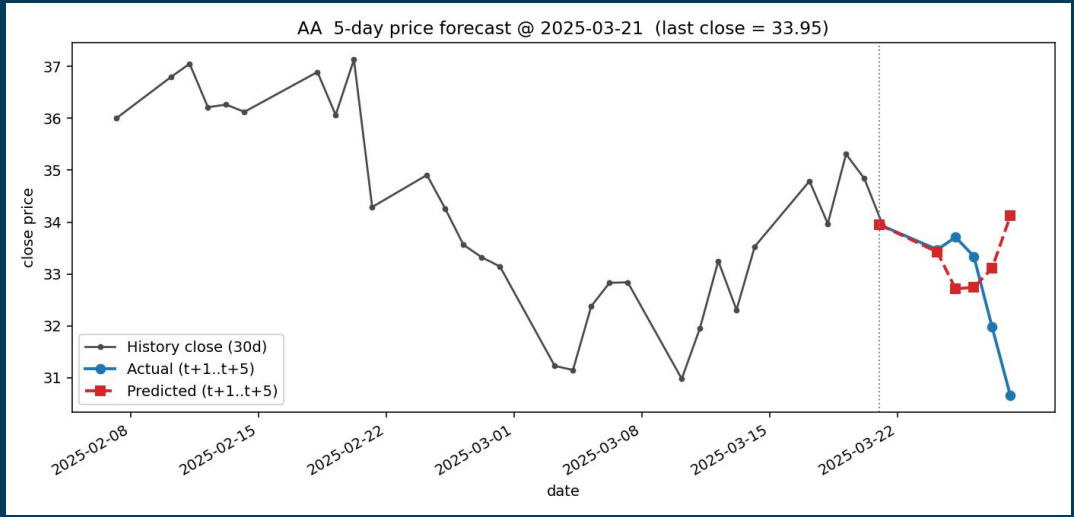
TSMixer

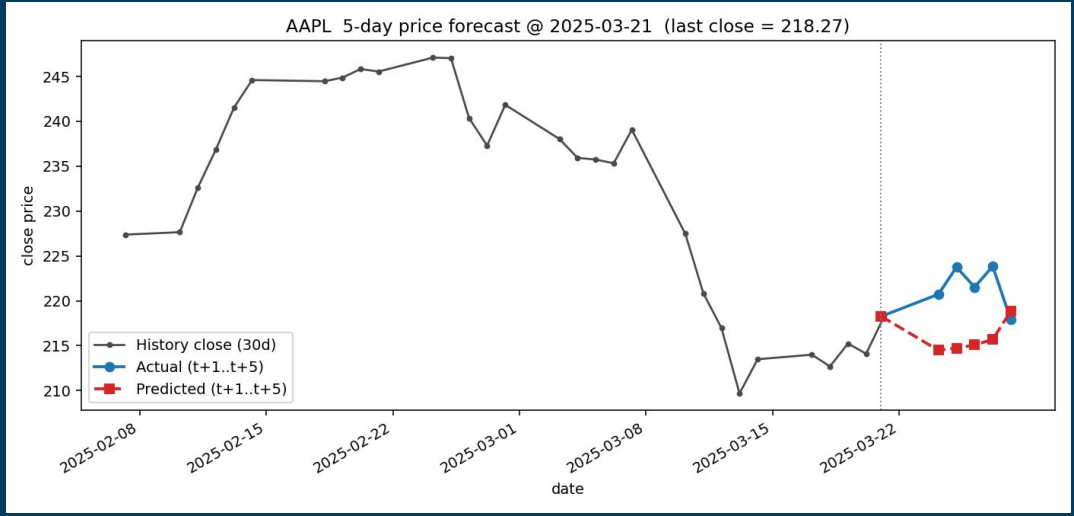
1. Features: Returns, price ranges, volume, volatility, RSI, ATR and more
2. Input/Output: 60-day window → 5-day forecast
3. Model: Mixer-based architecture (4 blocks: time + feature mixing)
4. Ticker embedding (stock-specific context)
5. Learnable recency weights
6. Direct multi-step forecasting



RESULTS

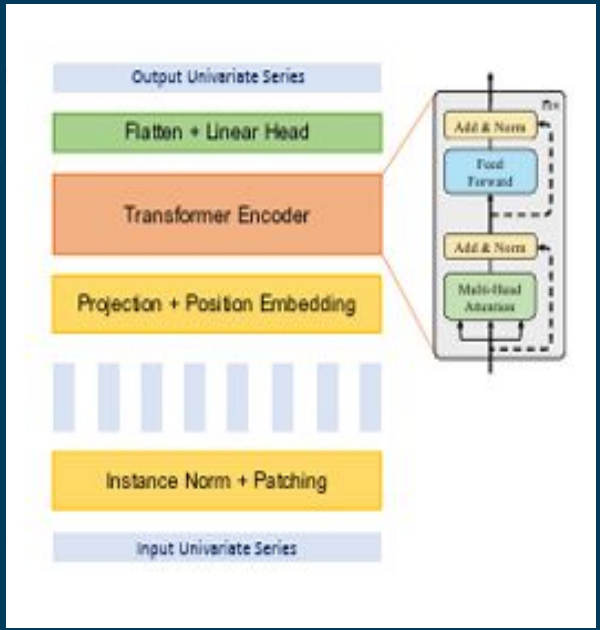
Metric	Value
MAE	0.1474
MSE	0.0496
RMSE	0.2228





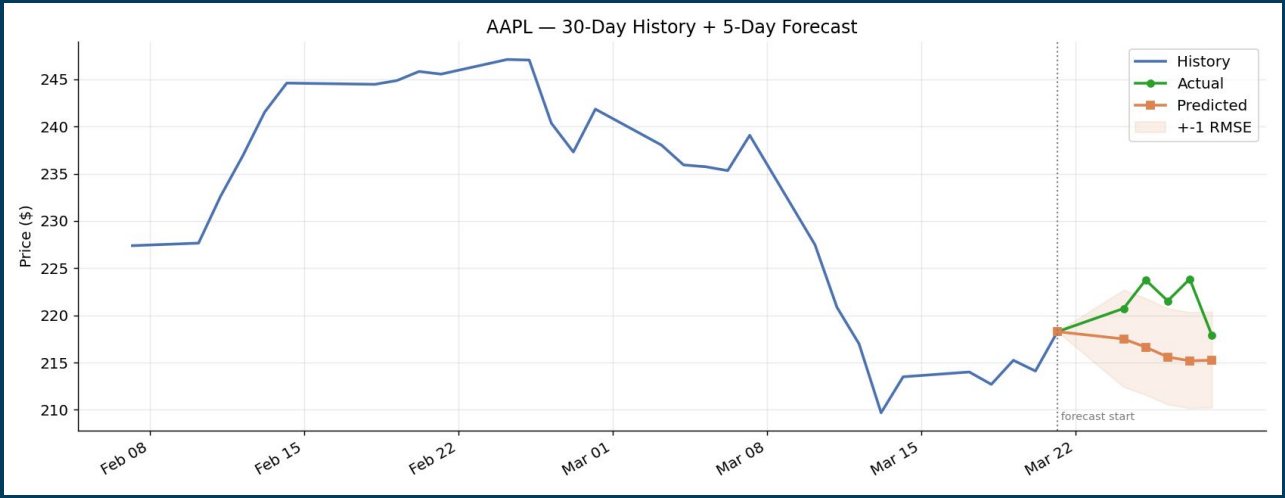
PatchTST

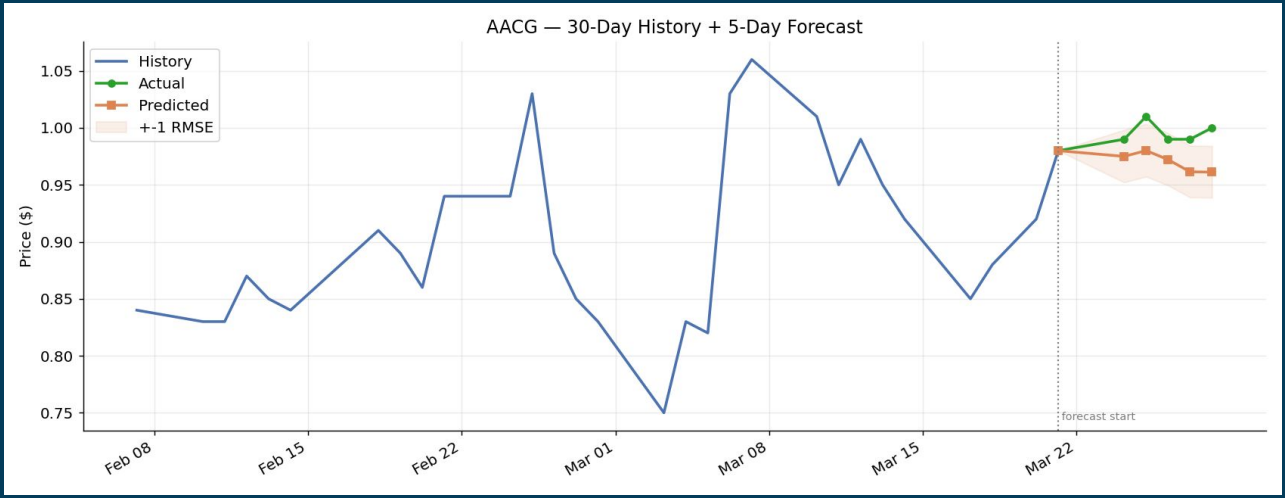
1. Training: AdamW + cosine LR, Huber loss, gradient clipping, early stopping
2. Channel-Independent Patching: Each feature processed separately → overlapping patches
3. Patch Encoding: 128-dim embeddings → 3 Transformer blocks (8 heads, ~410K params)
4. RevIN: Normalize per channel, revert after prediction
5. Output Head: Target-only (5-day forecast)



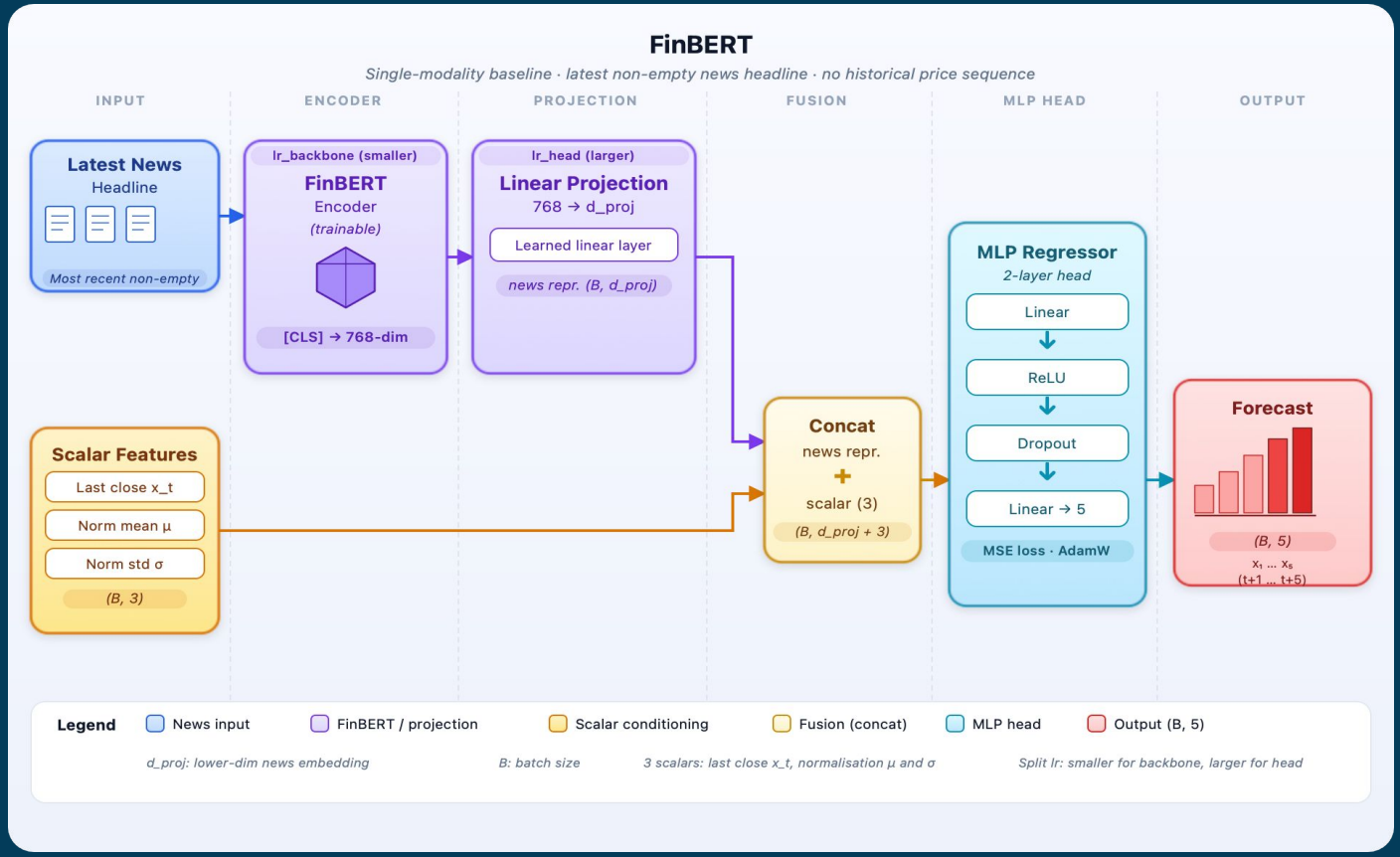
RESULTS

Metric	Value
MAE	0.0543
MSE	0.1466
RMSE	0.233



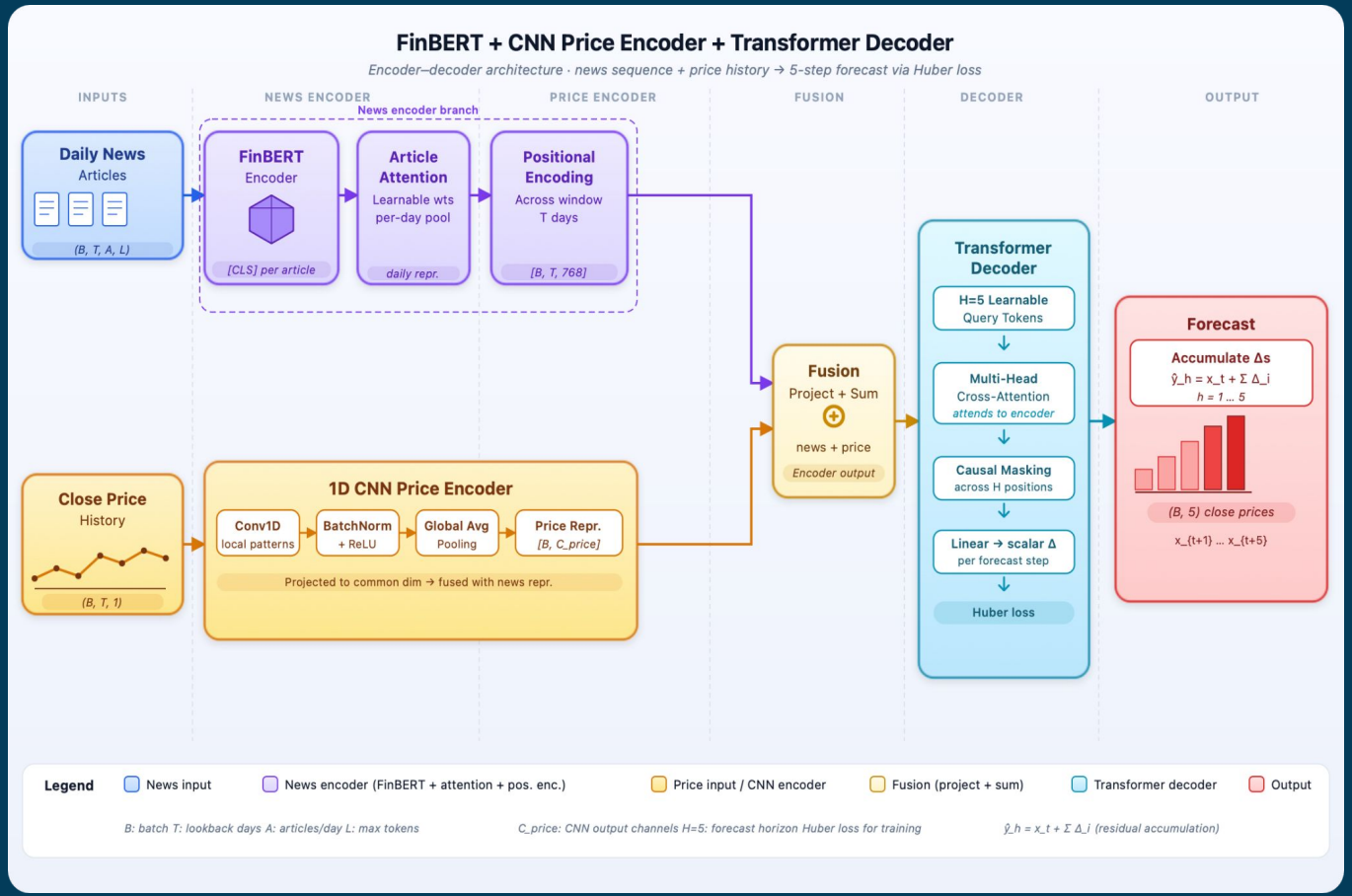


Multimodal: Finbert + MLP



Metric	Value
MAE	0.1602
MSE	0.0332
RMSE	0.1821

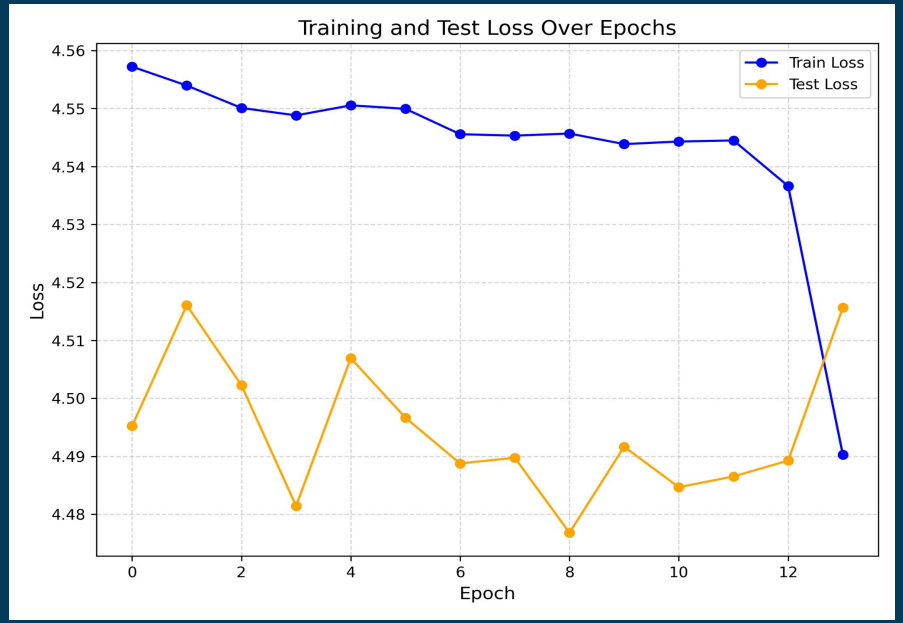
Multimodal: Finbert + CNN + transformer decoder

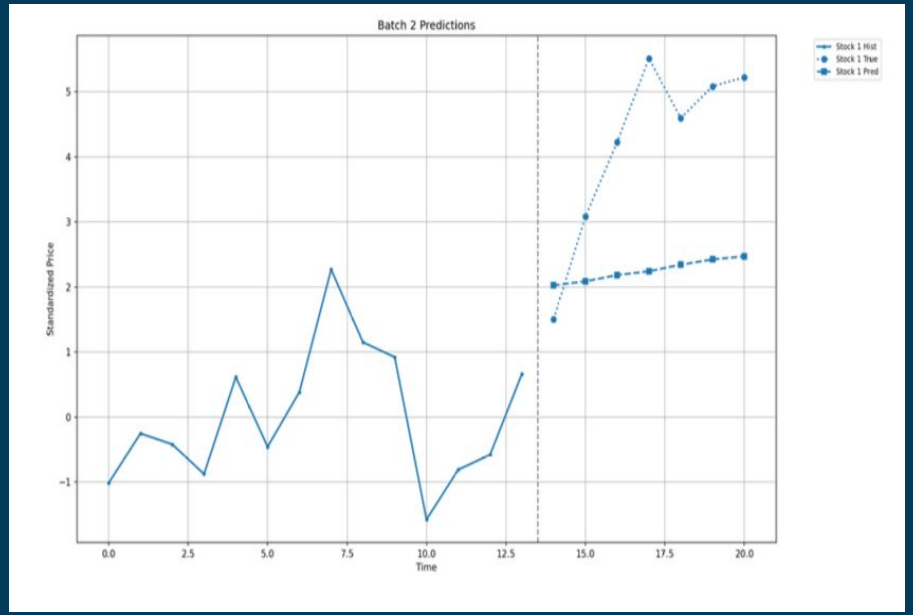


Metric	Value
MAE	0.0302
MSE	0.0009
RMSE	0.0316

Plots for FinBERT

Here are some plots for various experiments we did on FinBERT based models





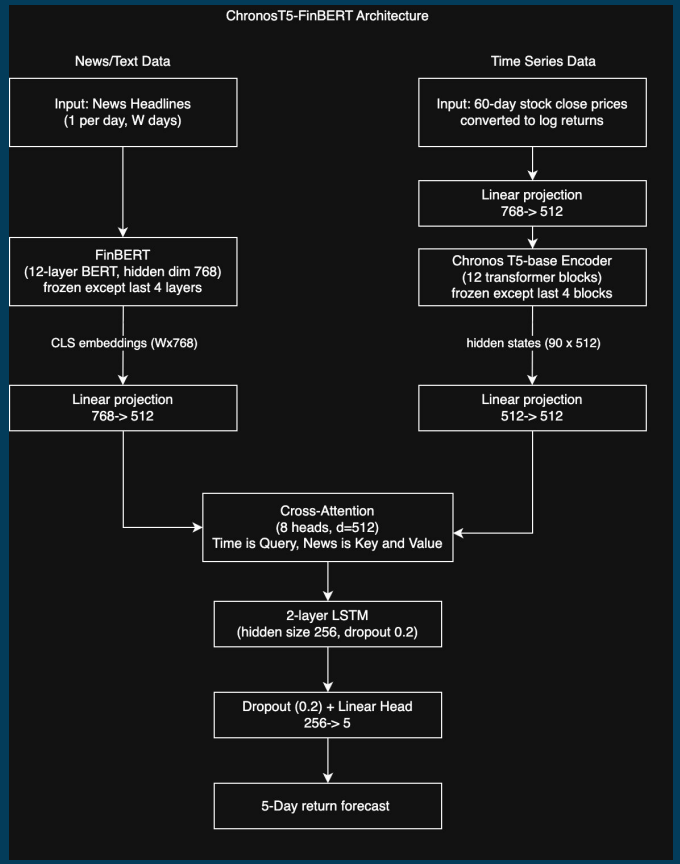
Multimodal Chronos T5 with FinBERT

Finbert : Preprocessing

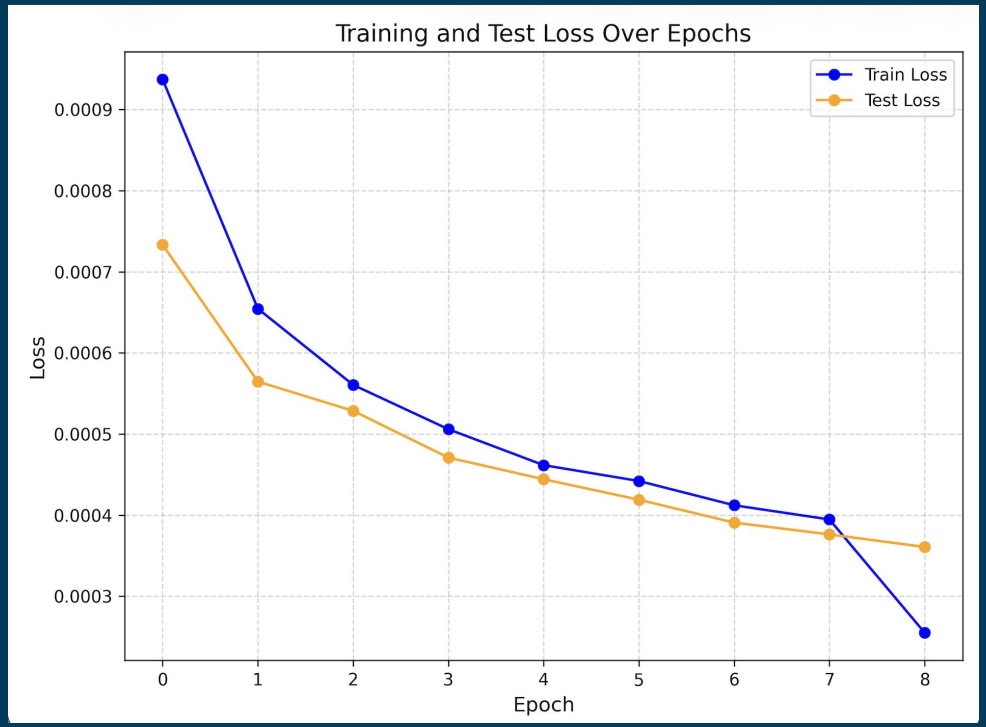
a. Text Processing - Financial news tokenized using FinBERT tokenizer with max length = 512 (padding + truncation), lowercase, punctuation removal

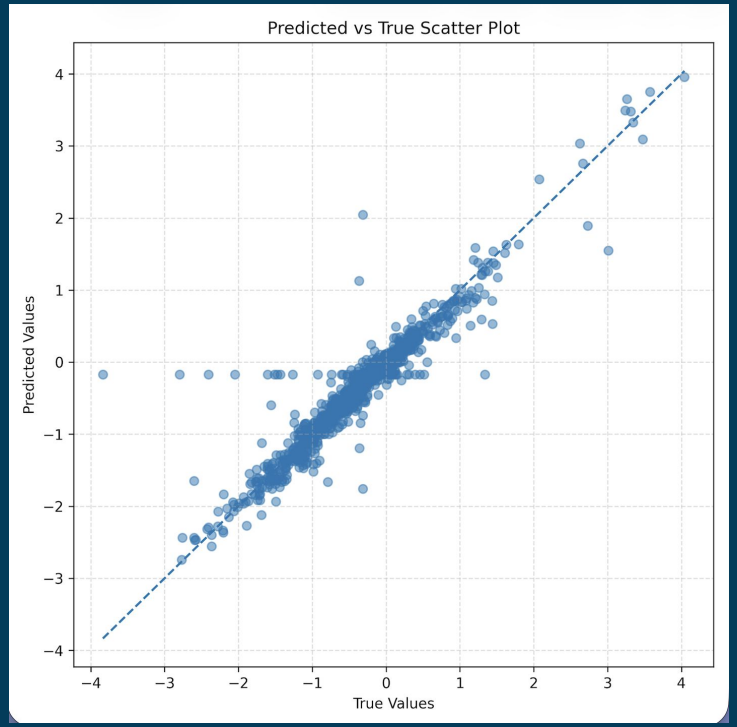
b. Missing Data Handling - Numerical values → mean imputation and no NaNs during training

c. Target Preparation - Targets normalized using input statistics, Maintains consistent scale between input/output



Results


Training and Test Loss Over Epochs
Train Loss
Test Loss
Loss
0.0009
0.0008
0.0007
0.0006
0.0005
0.0004
0.0003
0
1
2
3
4
5
6
7
8
Epoch


Predicted vs True Scatter Plot
Predicted Values
True Values
4
3
2
1
0
−1
−2
−3
−4

Chronos T5 - Finbert Multimodal

Metric	Chronos - FinBERT
Train MSE	0.016
Train RMSE	0.128
Train MAE	0.043
Test MSE	0.0156
Test RMSE	0.1249
Test MAE	0.0418

TFT-Finbert Multimodal

Preprocessing & Model Architecture

1. **Preprocessing**

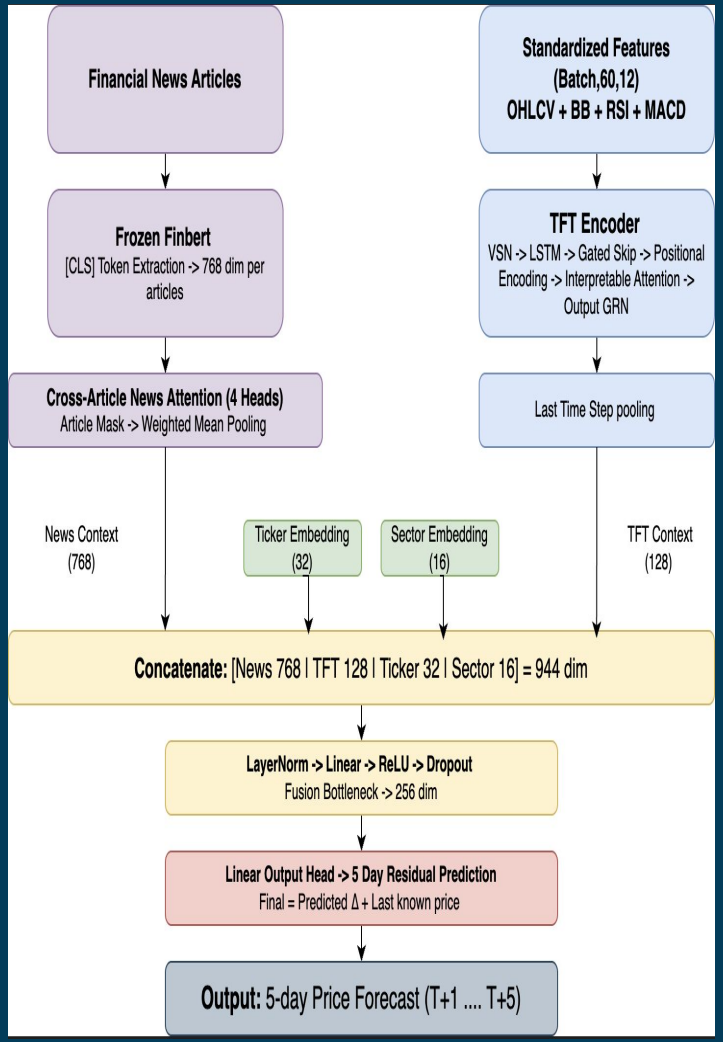
a. **News:** 1 article per day across 14-day window, tokenized via FinBERT tokenizer.

b. **Time series:** OHLCV + 7 technical indicators (Bollinger bands, RSI, MACD)

2. **Model**

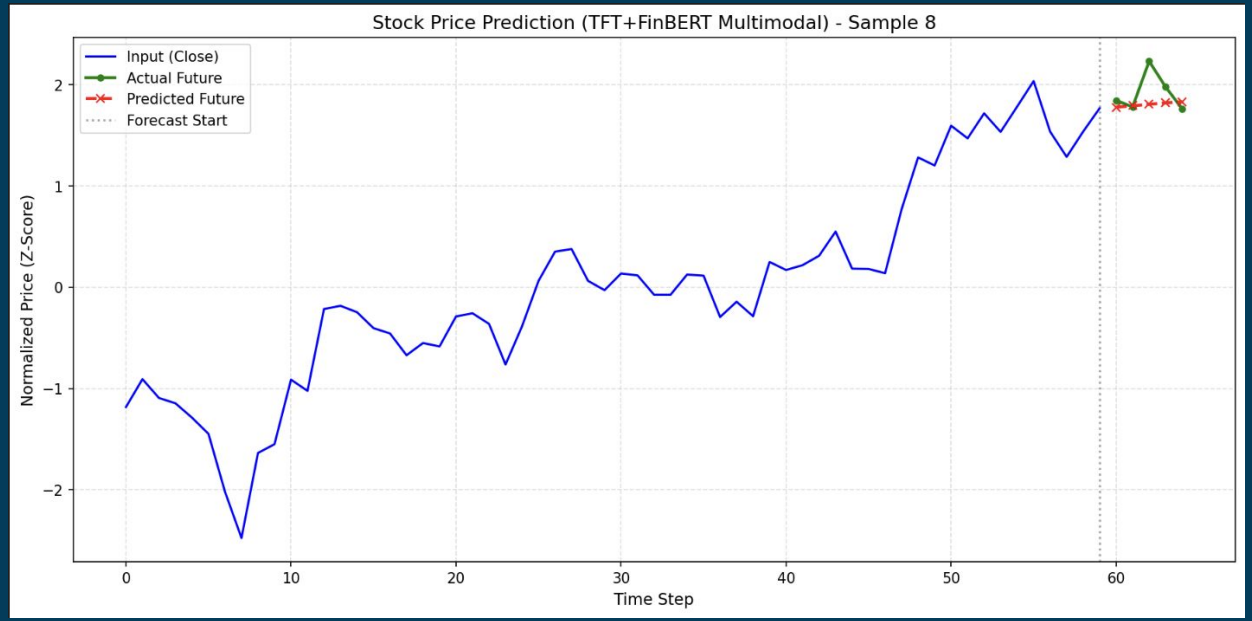
a. **Multimodal Encoding:** Combines news sentiment from FinBERT with temporal price patterns from the TFT.

b. **Integrated 5-Day Forecast:** Fuses all data streams into a single pass to predict future price trajectories.



TFT-FINBERT Multimodal Results

Metric	Value
MAE	0.03146
MSE	0.00146
RMSE	0.038212



Results

MODELS/METRICS	RMSE	MSE	MAE
LSTM	0.03794	0.00143	0.02939
PatchTST	0.233	0.0543	0.1466
TSMixer	0.2228	0.0496	0.1474
TFT	0.0681	0.00463	0.0358
TCN - Transformer	0.0673	0.00453	0.0355
FinBERT	0.1821	0.0332	0.1602
Chronos T5 - FinBERT	0.1249	0.0156	0.0418
TCN-FinBERT	**0.03029**	**0.00091**	**0.0286**
TFT-FinBERT	0.038212	0.00146	0.03146
FinBERT + CNN Encoder - Transformer Decoder	0.0316	0.00099	0.0302

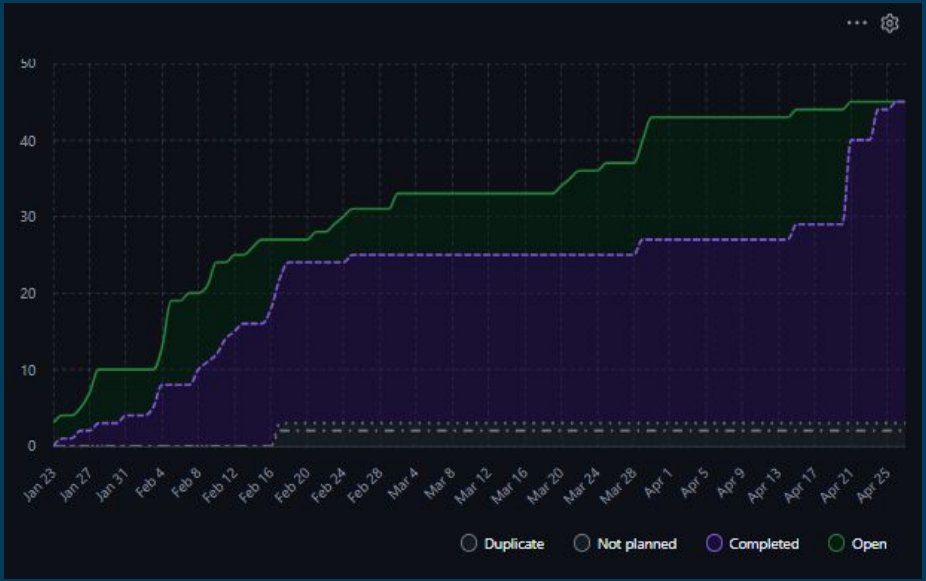
Conclusion

1. News Modality Adds Consistent Value Integrating financial news via FinBERT embeddings improved performance over every unimodal baseline — event-driven textual signals capture information that price history alone cannot recover.
2. Fusion Mechanism Matters More Than Temporal Encoder Choice Cross-attention and encoder-decoder fusion outperform naive concatenation. Dynamic interaction between text and price representations is key — not just having both modalities present.
3. Foundation Models Require Domain Adaptation Zero-shot Chronos models performed worst overall. Fine-tuning (ChronosFinBERT) partially recovered performance, but still trailed all fully supervised multimodal models.

Future Work

- Incorporate SEC XBRL Financial Reports Structured fundamental data (cash flow, equity, earnings) was collected and preprocessed but not used in any fusion model. A dedicated low-frequency encoder with learned temporal alignment to daily observations is the natural next step.
- Full Evaluation of Graph-Based Architecture (TGNN) Sector-level co-movement is well-documented in equity markets. The implemented Temporal Graph Neural Network was not evaluated due to resource constraints — this remains a high-priority item.
- Ablation of Individual Fusion Components Isolate the contribution of: cross-article self-attention vs. mean pooling, learned vs. fixed decoder query tokens, and number of fine-tuned FinBERT layers — currently these effects are entangled.
- Richer Evaluation Metrics Move beyond RMSE/MSE/MAE to include directional accuracy, Sharpe ratio simulation, and maximum drawdown analysis under a signal-following strategy.

Project Management



Research Paper

Multimodal Stock Forecasting with Financial News and Price Sequences: A Controlled Comparison of Fusion Architectures

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Abstract

Accurate stock forecasting depends on heterogeneous signals that range from raw market histories to unstructured financial news and structured corporate disclosures. This paper presents a standalone study of multimodal financial forecasting using the FinMultiTime benchmark. We evaluate price-only baselines, a text-only sentiment model, zero-shot time-series foundation models, and three strong multimodal price–news fusion architectures under a fixed chronological protocol on S&P 500 equities. To make the modeling contribution clearer, we formalize the forecasting task, describe the common news-encoding pipeline, and explain the design trade-offs behind TCN-FinBERT, TFT-FinBERT, and FinBERT+CNN+Transformer in greater detail. Across all evaluated systems, TCN-FinBERT achieves the lowest RMSE of 0.03029, followed by FinBERT+CNN+Transformer at 0.0316, while weaker late fusion and zero-shot foundation-model transfer lag behind. The results support a consistent conclusion: in multimodal financial forecasting, the mechanism used to align price dynamics with textual information matters at least as much as the temporal backbone itself. This paper therefore offers both an empirical benchmark and a practical design guide for future multimodal forecasting systems in finance.

1 Introduction

Financial time-series forecasting remains difficult because asset prices respond to multiple information channels simultaneously, including recent price dynamics, breaking news, investor sentiment, and firm-level fundamentals. Price-only models can capture local temporal regularities, but they often miss abrupt sentiment shifts and information shocks that arrive through text or disclosures [13, 15]. Recent multimodal benchmarks such as FinMultiTime make it possible to study these channels jointly by aligning time series, text, and structured financial data at scale [16].

Despite this progress, it is still unclear whether performance gains come primarily from richer data, from stronger model classes, or from the way modalities are fused. Many multimodal studies introduce a new architecture and a new data recipe at the same time, which makes attribution difficult [5, 11]. This study addresses that issue through a controlled benchmark on S&P 500 stocks: the prediction target, temporal split, and evaluation pipeline are held fixed while different unimodal and multimodal architectures are compared.

This paper makes three contributions. First, it provides a clear mathematical task definition for multimodal short-horizon stock forecasting. Second, it explains the three strongest multimodal models in more depth, including how each model encodes price histories, aggregates news, and maps fused representations to a five-day forecast. Third, it shows that the mechanism used to inject news into a temporal forecaster is the dominant design choice: structured fusion with FinBERT consistently improves over strong price-only baselines, whereas zero-shot foundation models and weak late fusion lag behind.

2 Problem Formulation

Let i index a stock and t index the last day in a lookback window. Each supervised example is a tuple

$$\mathcal{S}_{i,t} = \left(\mathbf{P}_{i,t}, \mathcal{N}_{i,t}, \mathbf{m}_i, \mathbf{y}_{i,t}\right),$$

where $\mathbf{P}_{i,t} \in \mathbb{R}^{T \times d_p}$ denotes the historical price-feature matrix, $\mathcal{N}_{i,t}$ denotes the aligned set of news articles in the window, $\mathbf{m}_i$ collects static metadata such as ticker and sector identity, and

$\mathbf{y}_{i,t} \in \mathbb{R}^H$ is the vector of future close prices. In this study, $H = 5$ trading days and $T \in \{30, 60, 90\}$ depending on the model family.

The forecasting function is written as

$$\hat{\mathbf{y}}_{i,t} = f_\theta(\mathbf{P}_{i,t}, \mathcal{N}_{i,t}, \mathbf{m}_i),$$

where f_θ may reduce to a price-only model or a text-only model when some inputs are omitted. Several architectures in this study forecast residual increments rather than prices directly. In that case, the model outputs $\Delta_{i,t,1:H}$ and reconstructs future closes as

$$\hat{x}_{i,t+h} = x_{i,t} + \sum_{u=1}^{h} \Delta_{i,t,u}, \qquad h = 1, \ldots, H.$$

This direct multi-step formulation avoids the error compounding that would arise if the next prediction were recursively fed back as input.

The primary evaluation metric is root mean squared error,

$$\mathrm{RMSE} = \sqrt{\frac{1}{NH} \sum_{i=1}^{N} \sum_{h=1}^{H} (y_{i,h} - \hat{y}_{i,h})^2},$$

reported together with MSE and MAE in our evaluation. We emphasize RMSE in this paper because it captures the overall model ranking most clearly across the benchmark.

3 Related Work

Classical and neural stock forecasting methods have long relied on historical market features. Recurrent architectures such as LSTMs remain important baselines for financial prediction because they can model sequential dependence without requiring extremely large datasets [6, 4]. More recent sequence models have shifted toward patch-based Transformers and parameter-efficient mixing architectures, including PatchTST and TSMixer, which improve long-horizon forecasting in many generic time-series settings [12, 3]. Temporal Fusion Transformers (TFTs) extend this line of work with variable selection, gating, and interpretable attention for multi-horizon prediction [9].

On the text side, domain-specific language models such as FinBERT provide representations of financial sentiment and narrative context that are better aligned with market language than generic BERT-style encoders [7]. Recent multimodal stock-prediction systems therefore combine textual encoders with temporal backbones, often through attention, gating, or graph-based fusion [5, 11]. The main open question is not whether text helps at all, but how to inject it

Table 1: FinMultiTime data summary used to define the project setting.

Dataset type	Size	Tickers	Frequency
Semi-annual reports	84.04 GB	2,676	Semi-annual
Stock time series	1.83 GB	4,213	Daily
Daily text	14.10 GB	4,694	Daily
Total	108.64 GB	—	—

into a temporal forecaster without allowing the high-dimensional text embedding to dominate the price signal.

Time-series foundation models such as Chronos and Chronos-2 have raised the possibility of zero-shot or lightly adapted forecasting across domains [2, 1]. However, evidence on financial equities remains mixed, with domain mismatch often leading to underperformance relative to supervised in-domain baselines [14]. This setting is therefore especially useful because it compares conventional supervised models, domain-specific language models, and foundation-model variants under one evaluation protocol.

4 Data and Preprocessing

The experiments are based on the FinMultiTime dataset [16]. The original benchmark spans multiple modalities and markets from 2000 to 2025, while this study restricts attention to S&P 500 equities and excludes the image modality in order to keep the comparison centered on price histories, news, and structured reports. All experiments use a chronological 70%/15%/15% train/validation/test split to avoid look-ahead bias.

Price inputs include OHLCV variables and technical indicators such as RSI, MACD, and Bollinger Bands. The study uses normalized per-stock features to improve stationarity and comparability across assets. News items are aligned to daily timestamps and either summarized into scalar sentiment labels or encoded directly by FinBERT. For the multimodal models emphasized here, the richer FinBERT embeddings are the main representation, because they preserve polarity, intensity, and contextual nuance better than a coarse ordinal label.

The preprocessing pipeline also constructs ticker and GICS-sector metadata. These variables are important because they provide static contextual information that price and news alone do not capture. Structured SEC-derived indicators were prepared during preprocessing, but they were not fully integrated into the three strongest quantitative models; we therefore treat them as an important direction for future work

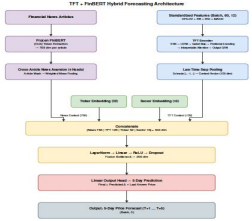

Figure 2: TFT-FinBERT late-fusion architecture.

5.4 FinBERT + CNN Price Encoder + Transformer Decoder

The FinBERT+CNN+Transformer model uses a richer encoder–decoder design. On the news side, daily articles are independently encoded by FinBERT and aggregated through learnable article attention within each trading day; positional encodings then create a time-indexed sequence of daily news embeddings. On the price side, the close-price history is passed through a one-dimensional CNN with convolution, batch normalization, ReLU, and global average pooling to form a compact price representation. The CNN branch is intentionally shallow, emphasizing local price-shape extraction rather than deep temporal compression.

Both streams are projected into a common latent space and summed to form the encoder memory. A Transformer decoder with $H = 5$ learnable query tokens attends to this memory and outputs residual price increments that are accumulated onto the last observed close. If $\mathbf{M}$ denotes the fused encoder memory and $\mathbf{q}_h$ denotes the query for horizon h, each step prediction is obtained through cross-attention followed by a scalar projection. This model is therefore the most explicit cross-modal interaction design in the benchmark: price and news are aligned before the horizon-specific decoder generates outputs.

5.5 Training Objective and Design Comparison

Most high-performing models in this study optimize a robust Huber objective rather than plain squared error. For a target y and prediction $\hat{y}$,

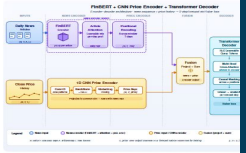

Figure 3: FinBERT+CNN+Transf
coder architecture.

Table 2: Comparison of the three h
multimodal models.

Model	Price encoder	Fusion strategy
TCN-FinBERT	Dilated TCN + Transformer	Late fusion after strong temporal encoding
TFT-FinBERT	TFT with variable selection and attention	Late fusion after interpretable temporal compression
FinBERT+CNN+Transformer	1D CNN price encoder + Transformer decoder	Encoder–decoder fusion with horizon queries

the scalar Huber loss is

$$\mathcal{L}_\delta(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2, & |y - \\ \delta|y - \hat{y}| - \frac{1}{2}\delta^2, & \text{othe} \end{cases}$$

This is appealing in equities becaus
market moves are common and can
pure MSE training. In practical terms
main multimodal models differ less i
than in where the fusion occurs and
temporal structure is preserved befor

6 Experimental Setup

All models are trained or evaluated o
chronological split and forecast the
trading days. Hyperparameters are
dependently for each architecture
dation RMSE, and the best validati
uration is carried into test evaluat
choice should be interpreted as a best
comparison rather than a shared-gri
The model zoo includes price-only
(LSTM, PatchTST, TSMixer, TFT,
Transformer), a text-only FinBERT
zero-shot Chronos variants, a partial
Chronos-FinBERT hybrid, and the t
modal models highlighted above.

A useful design feature shared by
lines is direct multi-step forecasting
of rolling out predictions autoregres

DEMO

CONFIGURATION

MODEL

LSTM

TICKER

MSFT

RUN BACKTEST

weights loaded.

STOCK BACKTESTING TERMINAL

MSFT · LSTM BACKTEST

SESSION 2026-04-27 17:43 UTC

Price Apr 20	Actual Close	Forecast Close
$418.07	$424.82	$425.21
	↑ +1.61%	↑ +1.71%

BACKTEST DATA

BACKTEST DATA

Date	Actual	Predicted	Abs Error
2026-04-21	$424.16	$423.77	$0.39
2026-04-22	$432.92	$417.37	$15.55
2026-04-23	$415.75	$424.73	$8.98
2026-04-24	$424.62	$431.09	$6.47
2026-04-27	$424.82	$425.21	$0.39

THANK YOU!